

POWERFACTORY

PowerFactory 2024

Technical Reference

Voltage Transformer

StaVt, StaVtsec, TypVt, TypVtsec

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

Copyright © 2024 DIgSILENT GmbH
All rights reserved. No part of this publication may be reproduced or distributed in any form without written permission of DIgSILENT GmbH.

March 28, 2024
PowerFactory 2024
Revision 2

Contents

1 General Description **1**

1.1 Voltage Transformer Type 1

1.2 Secondary Voltage Transformer Type 1

1.3 Voltage Transformer 2

1.4 Secondary Voltage Transformer Winding 4

2 Calculation Models **5**

2.1 Calculation of Primary Input 5

2.2 Ideal 2- or 3-Phase Voltage Transformer 6

2.2.1 Load Flow, Short-Circuit Calculation and RMS Simulation 6

2.2.2 EMT Simulation 7

2.3 Detailed 2- or 3-Phase Voltage Transformer 9

2.3.1 Load Flow, Short-Circuit Calculation and RMS Simulation 9

2.3.2 EMT Simulation 13

2.4 Capacitive 2- or 3-Phase Voltage Transformer 18

2.4.1 Load Flow, Short-Circuit Calculation and RMS Simulation 18

2.4.2 EMT Simulation 24

2.5 Ideal Single-Phase Voltage Transformer 30

2.5.1 Load Flow, Short-Circuit Calculation and RMS Simulation 30

2.5.2 EMT Simulation 31

2.6 Saturation Characteristic 33

2.6.1 Piecewise linear saturation 33

2.6.2 Polynomial saturation 33

3 References **35**

List of Figures **36**

List of Tables **37**

1 General Description

This document describes the *PowerFactory* voltage transformers and their calculation models.

1.1 Voltage Transformer Type

The voltage transformer type (*TypVt*) facilitates the input of data for the primary side of the transformer. It can be used for generic voltage transformers or to replicate a specific data sheet.

The *Basic Data* page offers the possibility to input multiple primary taps and select the level of detail to be used.

Note: For capacitive voltage transformers, only one primary voltage can be entered.

The *Additional Data* page offers input for the parameters of a detailed, inductive voltage transformer. The *Additional CVT Data* page extends the input of the *Additional Data* page with the input for a detailed, capacitive voltage transformer.

Note: The *Additional Data* and *Additional CVT Data* pages only require input if the model type has been set accordingly.

If the input option for the *Rated Burden* is set to “Impedance”, the rated apparent power S_r is calculated as follows:

$$S_r = \frac{{U_{r,s}}^2}{Z_b} \tag{1}$$

where:

$U_{r,s}$ is the rated secondary voltage in [V]

Z_b is the rated burden in [Ω] (*Zb*)

As the type itself does not specify a rated secondary voltage, the secondary tap set in the corresponding element is used when calculating the rated power for calculation models.

1.2 Secondary Voltage Transformer Type

The secondary voltage transformer type (*TypVtsec*) allows input of data for the secondary winding of the transformer. The *Basic Data* page offers input for multiple secondary taps. Otherwise similar to the detailed *TypVt* in terms of data input, the *Advanced Data* page is limited to parameters relevant to a secondary winding.

Note: Whether or not the parameters on the *Advanced data* page influence the calculation will ultimately depend on the element configuration.

The calculation of the rated apparent power S_r is identical to *TypVt* (see Section 1.1), but will use the first entry of the secondary tap list for $U_{r,s}$ while editing the type.

1.3 Voltage Transformer

A voltage transformer (*StaVt*) can be defined by a *TypVt* and an optional *TypVtsec*. It is also possible to assign no type at all, in which case the voltage transformer is considered as an ideal transformer with a $1:1$ ratio.

Measurement Location:
Typically a voltage transformer is placed inside a cubicle (*StaCubic*) within the network model to indicate the measurement location. For advanced applications, it is also possible to specify the measurement location directly via the "Reference (*pbusbar*)" selection.

The following measurement locations can be selected:

- Cubicle (*StaCubic*) : measure the voltage at the cubicle's parent terminal
- Terminal (*ElmTerm*) : measures the voltage at the terminal
- Voltage Transformer (*StaVt*, *StaVtsec*) : measure as an an auxiliary transformer to the selected transformer

---

Note: Voltage transformers can also be used to measure the voltage at DC equipment. In this case the transformer will always be treated as ideal single-phase transformer.

Tap Selection:
If multiple taps are defined in the associated *TypVt*, the primary tap to be used can be selected here. If multiple primary taps are available, the "Detect primary tap" button can be used to automatically select the tap, which is closest to the nominal voltage of the measured terminal.

If no *TypVtsec* is assigned, the secondary tap to be used can be selected from a list of values commonly found in voltage transformers. If a *TypVtsec* is assigned, the selectable values are taken from the type.

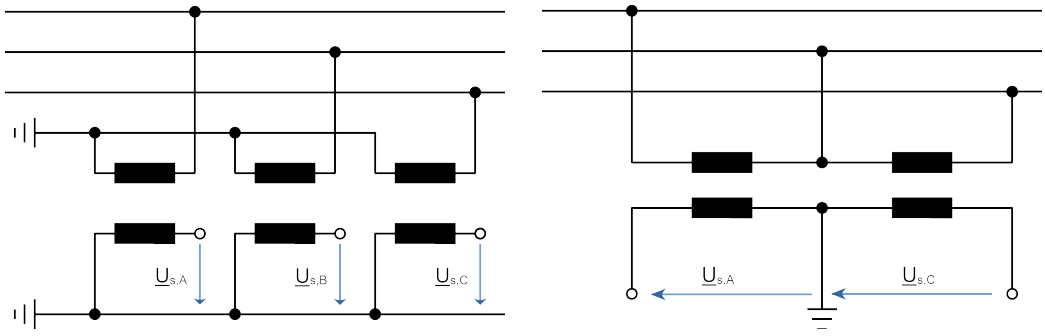
Note: The taps are always in line-line quantities, even if the transformer is in *wye* connection. The conversion from line-line to line-ground quantities if performed internally where necessary.

Connection:
The voltage transformer element can be configured to measure 1-3 phases. In the case of a 3-phase voltage transformer, the vector group and phase rotation for the connection of the secondary side can be configured. The *wye* ("Y") and *open delta* ("V") connections are shown in Figures 1.1a and 1.1b respectively.

Note: Even though an open delta connection only uses 2 voltage transformers, it is classified as 3-phase, because it implicitly provides three line-line voltages.

For single-phase transformers one additional phase, 'N', can be selected. A single-phase transformer with phase 'N' selected models a *broken delta* winding (see Figure 1.2) which is used to measure the zero sequence displacement voltage.

For 2-phase current transformers the two measured phases, which are always *wye* connected, can be selected.



(a) Wye connection (Y)

(b) Open delta connection (V)

Figure 1.1: 3-phase voltage transformer connections

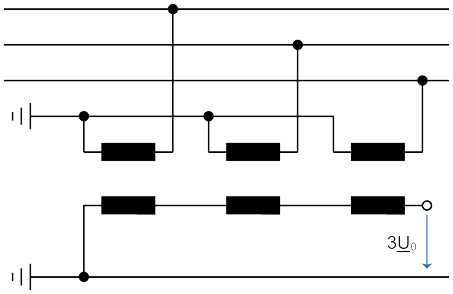


Figure 1.2: 1-phase voltage transformer in broken delta connection (N)

Additional Windings:
It is possible to group several secondary voltage transformer windings (see Section 1.4) in one voltage transformer. They can be created and accessed with the buttons “Add Winding” and “Edit Windings”. The contained windings are listed with their nameplate.

Additional Data:
The *Additional Data* page can be used to enable and configure the detailed model for calculation. For detailed information, please see Sections 2.3 and 2.4. Only 2- or 3-phase voltage transformers can be set to detailed. Additionally, the associated type must be configured for either detailed or capacitive voltage transformer input (see Section 1.1).

If the input option for the *Actual Burden* is set to “Impedance” or “Apparent Power”, the parameters R_b and $Lb$are calculated as follows:

Impedance:

$$R_b = Z_b * \cos\varphi_b \tag{2}$$

$$L_B = \frac{Z_b * \sqrt{1-(\cos\varphi_b)^2}}{2 \cdot \pi \cdot f_N} \tag{3}$$

Apparent Power:

$$R_b = \frac{U_{tap,s}^2}{S_b} * \cos\varphi_b \tag{4}$$

$$L_B = \frac{\frac{U_{tap,s}^2}{S_b} * \sqrt{1-(\cos\varphi_b)^2}}{2 \cdot \pi \cdot f_N} \tag{5}$$

where:

$U_{tap,s}$ is the selected secondary voltage in [V] (*stapset*)

Z_b is the burden impedance in [Ω] (*Zburd*)

S_b is the burden apparent power in [VA] (*Sburd*)

$\cos\varphi_b$ is the burden power factor (*cosburd*)

f_N is the nominal frequency of the network in [Hz]

1.4 Secondary Voltage Transformer Winding

A secondary voltage transformer winding (*StaVtsec*) can either be placed inside a voltage transformer or a combined instrument transformer (*StaCombi*). The configuration options are very similar to voltage transformers (see Section 1.3), but reduced to the secondary side. The measurement location is always derived from the parent transformer. Additionally, for 3-phase secondary voltage transformer windings, the connection is fixed to the one used by the parent transformer. Whether a detailed model is configurable or not depends on the type configuration of the parent transformer.

Note: Apart from sharing the same measurement location and certain configuration options with their parent transformer, secondary voltage transformers windings are completely separate models. During calculations, there is no influence of a secondary transformer winding on its parent (or vice versa) or on any other secondary transformer windings in the same transformer.

---

2 Calculation Models

This section describes the calculation models.

2.1 Calculation of Primary Input

Typically the input signals for primary voltage are automatically connected to the corresponding measurement location. The per-unit values obtained in this manner are subsequently converted into primary quantities using the nominal voltage of the measurement location. In contrast to most *PowerFactory* models, voltage transformers always provide an unbalanced output, even in balanced calculations. The following equations are used to generate the necessary primary phase voltages from balanced calculations:

$$\underline{U}_{p,A} = \underline{U}_p \qquad \underline{U}_{p,B} = a \cdot \underline{U}_p \qquad \underline{U}_{p,C} = a^2 \cdot \underline{U}_p \tag{6}$$

where:

$\underline{U}_p$ is the symmetrical primary voltage in [V]

$\underline{U}_{p,X}$ is the generated primary voltage, phase X in [V]

a is the phasor rotation operator $a = e^{j120°}$

2.2 Ideal 2- or 3-Phase Voltage Transformer

2.2.1 Load Flow, Short-Circuit Calculation and RMS Simulation

***Wye*-connected voltage transformer:**

$$\underline{U}_{s,A} = \frac{1}{ratio} \cdot \underline{U}_{p,A} \tag{7}$$

$$\underline{U}_{s,B} = \frac{1}{ratio} \cdot \underline{U}_{p,B} \tag{8}$$

$$\underline{U}_{s,C} = \frac{1}{ratio} \cdot \underline{U}_{p,C} \tag{9}$$

***Open delta*-connected voltage transformer:**

$$\underline{U}_{s,A} = \frac{1}{ratio} \cdot (\underline{U}_{p,A} - \underline{U}_{p,B}) \tag{10}$$

$$\underline{U}_{s,B} = 0 \tag{11}$$

$$\underline{U}_{s,C} = \frac{1}{ratio} \cdot (\underline{U}_{p,C} - \underline{U}_{p,B}) \tag{12}$$

where:

$\underline{U}_{p,X}$ is the primary voltage, phase X in [V] (see Section 2.1)

$\underline{U}_{s,X}$ is the secondary voltage, phase X in [V] (*s:U2r_X* $+ j \cdot$ *s:U2i_X*)

$ratio$ is the transmission ratio (*c:ratio*), where:

$$ratio = \frac{U_{tap,p}}{U_{tap,s}}$$

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

For 2-phase, *wye*-connected voltage transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Table 2.1: Signals, ideal 2- or 3-phase VT, balanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ur_p	Primary Voltage, Real Part	p.u.	IN
Ui_p	Primary Voltage, Imaginary Part	p.u.	IN
U2r_A	Secondary Voltage, Real Part A	V	OUT
U2i_A	Secondary Voltage, Imaginary Part A	V	OUT
U2r_B	Secondary Voltage, Real Part B	V	OUT
U2i_B	Secondary Voltage, Imaginary Part B	V	OUT
U2r_C	Secondary Voltage, Real Part C	V	OUT
U2i_C	Secondary Voltage, Imaginary Part C	V	OUT

Table 2.2: Signals, ideal 2- or 3-phase VT, unbalanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ur_A	Primary Voltage, Real Part A	p.u.	IN
Ui_A	Primary Voltage, Imaginary Part A	p.u.	IN
Ur_B	Primary Voltage, Real Part B	p.u.	IN
Ui_B	Primary Voltage, Imaginary Part B	p.u.	IN
Ur_C	Primary Voltage, Real Part C	p.u.	IN
Ui_C	Primary Voltage, Imaginary Part C	p.u.	IN
U2r_A	Secondary Voltage, Real Part A	V	OUT
U2i_A	Secondary Voltage, Imaginary Part A	V	OUT
U2r_B	Secondary Voltage, Real Part B	V	OUT
U2i_B	Secondary Voltage, Imaginary Part B	V	OUT
U2r_C	Secondary Voltage, Real Part C	V	OUT
U2i_C	Secondary Voltage, Imaginary Part C	V	OUT

Table 2.3: Calculation parameters, ideal 2- or 3-phase VT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Usec:A	Secondary voltage A	V
Usec:B	Secondary voltage B	V
Usec:C	Secondary voltage C	V

The calculation parameters *c:Usec:A*, *c:Usec:B* and *c:Usec:C* show the magnitudes of the corresponding complex secondary voltages.

2.2.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation.

***Wye*-connected voltage transformer:**

$$U_{s,A} = \frac{1}{ratio} \cdot U_{p,A} \tag{13}$$

$$U_{s,B} = \frac{1}{ratio} \cdot U_{p,B} \tag{14}$$

$$U_{s,C} = \frac{1}{ratio} \cdot U_{p,C} \tag{15}$$

***Open delta*-connected voltage transformer:**

$$U_{s,A} = \frac{1}{ratio} \cdot (U_{p,A} - U_{p,B}) \tag{16}$$

$$U_{s,B} = 0 \tag{17}$$

$$U_{s,C} = \frac{1}{ratio} \cdot (U_{p,C} - U_{p,B}) \tag{18}$$

where:

$U_{p,X}$ is the primary voltage, phase X in [V] (see Section 2.1)

$U_{s,X}$ is the secondary voltage, phase X in [V] (*s:U2r_X*)

$ratio$ is the transmission ratio (*c:ratio*), where:

$$ratio = \frac{U_{tap,p}}{U_{tap,s}}$$

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

For 2-phase, *wye*-connected voltage transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Table 2.4: Signals, ideal 2- or 3-phase VT, EMT simulation

Signal	Description	Unit	Type
U_A	Primary Voltage A	p.u.	IN
U_B	Primary Voltage B	p.u.	IN
U_C	Primary Voltage C	p.u.	IN
U2r_A	Secondary Voltage A	V	OUT
U2r_B	Secondary Voltage B	V	OUT
U2r_C	Secondary Voltage C	V	OUT

Table 2.5: Calculation parameters, ideal 2- or 3-phase VT, EMT simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Usec:A	Secondary voltage A	V
Usec:B	Secondary voltage B	V
Usec:C	Secondary voltage C	V

The calculation parameters *c:Usec:A*, *c:Usec:B* and *c:Usec:C* show the magnitudes of the corresponding secondary voltages.

2.3 Detailed 2- or 3-Phase Voltage Transformer

Detailed voltage transformer models consider the influence of leakage and magnetising current to the resulting voltage measurement.

2.3.1 Load Flow, Short-Circuit Calculation and RMS Simulation

Saturation is considered with a steady state equivalent. The equivalent circuit is shown in Figure 2.1.

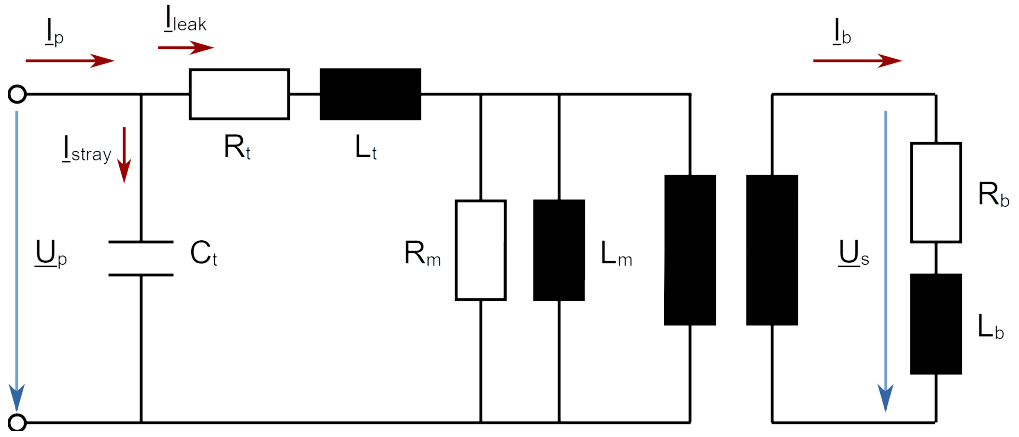


Figure 2.1: Equivalent circuit for a detailed VT used in load flow, short-circuit calculation and RMS simulation

Some of the equations below are given in *per-unit* for readability. The relevant system bases are calculated as follows:

$$Z_{base,p} = \frac{U_{tap,p}^2}{S_r} \qquad I_{base,p} = \frac{S_r}{\sqrt{3} \cdot U_{tap,p}} \qquad U_{base,p} = \frac{U_{tap,p}}{\sqrt{3}} \tag{19}$$

$$Z_{base,s} = \frac{U_{tap,s}^2}{S_r} \qquad I_{base,s} = \frac{S_r}{\sqrt{3} \cdot U_{tap,s}} \qquad U_{base,s} = \frac{U_{tap,s}}{\sqrt{3}} \tag{20}$$

where:

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

S_r is the rated apparent power S_r in [VA] (*t:Snom*)

Various equivalent impedances, derived from Figure 2.1 and used in the equations for output signals and calculation parameters, are calculated as follows:

Complex impedances:

$$\underline{z}_b = (R_b + j \cdot X_b) \cdot \frac{1}{Z_{base,s}} \tag{21}$$

$$\underline{z}_m = \frac{R_m \cdot j \cdot X_m}{R_m + j \cdot X_m} \cdot \frac{1}{Z_{base,p}} \tag{22}$$

$$\underline{z}_t = R_t + j \cdot X_t \cdot \frac{1}{Z_{base,p}} \tag{23}$$

$$\underline{z}_C = j \cdot X_C \cdot \frac{1}{Z_{base,p}} \tag{24}$$

Equivalent impedances:

$$\underline{z}_{b\|m} = \frac{\underline{z}_m \cdot \underline{z}_b}{\underline{z}_m + \underline{z}_b} \tag{25}$$

$$\underline{k}_{ind} = \frac{\underline{z}_{b\|m}}{\underline{z}_t + \underline{z}_{b\|m}} \tag{26}$$

$$\underline{z}_e = \frac{\underline{z}_C \cdot (\underline{z}_t + \underline{z}_{b\|m})}{\underline{z}_C + (\underline{z}_t + \underline{z}_{b\|m})} \tag{27}$$

where:

R_b is the actual burden resistance in [Ω] (*e:Rb*)

X_b is the actual burden reactance in [Ω], where:

$$X_b = \omega_N \cdot L_b$$

L_b is the actual burden inductance in [H] (*e:Lb*)

R_m is the magnetising resistance in [Ω] (*t:Rm*)

X_m is the magnetising reactance in [Ω], where:

$$X_m = \omega_N \cdot L_m$$

L_b is the magnetising inductance in [H] (*t:Lm*)

R_t is the leakage resistance in [Ω] (*t:Rt*)

X_t is the leakage reactance in [Ω], where:

$$X_t = \omega_N \cdot L_t$$

L_t is the leakage inductance in [H] (*t:Lt*)

X_C is the stray reactance in [Ω], where:

$$X_C = -\frac{1}{\omega_N \cdot C_t \cdot 10^{-9}}$$

C_t is the stray capacitance in [nF] (*e:Ct*)

$\underline{k}_{ind}$ is the complex factor for the inductive voltage divider

ω_N is the nominal angular velocity in [rad/s], where:

$$\omega_N = 2 \cdot \pi \cdot f_N$$

f_N is the nominal frequency of the network in [Hz]

***Wye*-connected voltage transformer:**

$$\underline{U}_{s,A} = \underline{U}_{p,A} \cdot \underline{k}_{ind} \cdot \frac{1}{ratio} \tag{28}$$

$$\underline{U}_{s,B} = \underline{U}_{p,B} \cdot \underline{k}_{ind} \cdot \frac{1}{ratio} \tag{29}$$

$$\underline{U}_{s,C} = \underline{U}_{p,C} \cdot \underline{k}_{ind} \cdot \frac{1}{ratio} \tag{30}$$

***Open delta*-connected voltage transformer:**

$$\underline{U}_{s,A} = (\underline{U}_{p,A} - \underline{U}_{p,B}) \cdot \underline{k}_{ind} \cdot \frac{1}{ratio} \tag{31}$$

$$\underline{U}_{s,B} = 0 \tag{32}$$

$$\underline{U}_{s,C} = (\underline{U}_{p,C} - \underline{U}_{p,B}) \cdot \underline{k}_{ind} \cdot \frac{1}{ratio} \tag{33}$$

where:

$\underline{U}_{p,X}$ is the primary voltage, phase X in [V] (see Section 2.1)

$\underline{U}_{s,X}$ is the secondary voltage, phase X in [V] (*s:U2r_X* $+ j \cdot$ *s:U2i_X*)

$ratio$ is the transmission ratio (*c:ratio*), where:

$$ratio = \frac{U_{tap,p}}{U_{tap,s}}$$

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

For 2-phase, *wye*-connected voltage transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Signal connections are identical to the ideal voltage transformer. See Tables 2.1 and 2.2.

Table 2.6: Calculation parameters, detailed 2- or 3-phase VT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Usec:A	Secondary voltage A	V
Usec:B	Secondary voltage B	V
Usec:C	Secondary voltage C	V
ip_A	Primary Current A	p.u.
ip_B	Primary Current B	p.u.
ip_C	Primary Current C	p.u.
ib_A	Burden Current A	p.u.
ib_B	Burden Current B	p.u.
ib_C	Burden Current C	p.u.
ileak_A	Leakage Current A	p.u.
ileak_B	Leakage Current B	p.u.
ileak_C	Leakage Current C	p.u.
ub_A	Burden Voltage A	p.u.
ub_B	Burden Voltage B	p.u.
ub_C	Burden Voltage C	p.u.
Zb:r	Burden, Impedance, Real Part	Ω
Zb:i	Burden, Impedance, Imaginary Part	Ω
Zb:phi	Burden, Impedance, Angle	°
Zb:mag	Burden, Impedance, Magnitude	Ω
Zb:phirad	Burden, Impedance, Angle	rad

The calculation parameters *c:Usec:A*, *c:Usec:B* and *c:Usec:C* show the magnitudes of the corresponding complex secondary voltages.

Note: *Per-unit* quantities are referred to the primary-side base system

---

The calculation parameters *c:ip_A*, *c:ip_B* and *c:ip_C* show the complex current magnitudes flowing into the whole voltage transformer and are calculated as follows:

$$i_p = \left| \frac{\underline{u}_p}{\underline{z}_e} \right| \tag{34}$$

The calculation parameters *c:ib_A*, *c:ib_B* and *c:ib_C* show the complex current magnitudes flowing into the burden and are calculated as follows:

$$i_b = \left| \frac{\underline{u}_p \cdot \underline{k}_{ind}}{\underline{z}_b} \right| \tag{35}$$

The calculation parameters *c:ileak_A*, *c:ileak_B* and *c:ileak_C* show the complex current magnitudes flowing into the leakage branch and are calculated as follows:

$$i_{leak} = \left| \frac{\underline{u}_p}{\underline{z}_t + \underline{z}_{b\|m}} \right| \tag{36}$$

The calculation parameters *c:ub_A*, *c:ub_B* and *c:ub_C* show the complex voltage magnitudes at the burden and are calculated as follows:

$$u_b = \left| \underline{u}_p \cdot \underline{k}_{ind} \right| \tag{37}$$

where:

$\underline{u}_p$ is connection dependent primary voltage in [p.u.]

2.3.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation. The saturation is considered with a non-linear magnetising inductance. The equivalent circuit is shown in Figure 2.2.

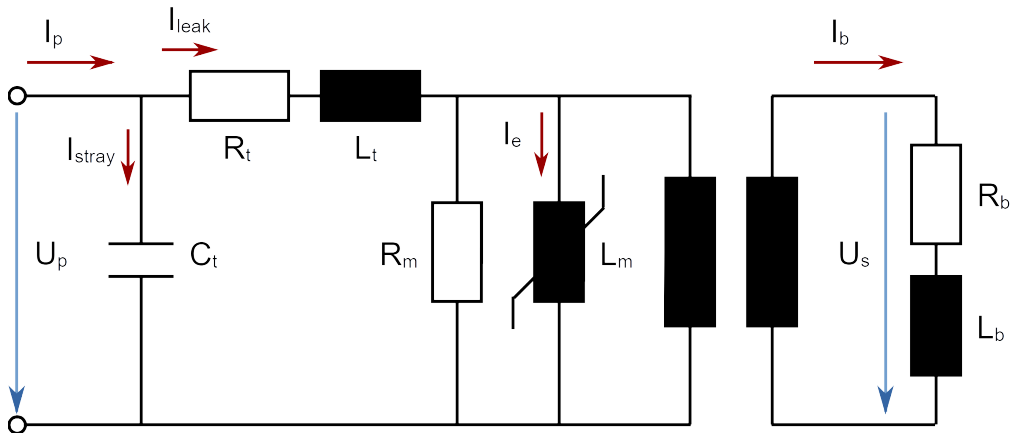


Figure 2.2: Equivalent circuit for a detailed VT used in EMT simulation

Some of the equations below are given in *per-unit* for readability. The relevant system bases are calculated as follows:

$$Z_{base,p} = \frac{U_{tap,p}^2}{S_r} \qquad I_{base,p} = \frac{\sqrt{2} \cdot S_r}{\sqrt{3} \cdot U_{tap,p}} \qquad U_{base,p} = \frac{\sqrt{2} \cdot U_{tap,p}}{\sqrt{3}} \tag{38}$$

$$Z_{base,s} = \frac{U_{tap,s}^2}{S_r} \qquad I_{base,s} = \frac{\sqrt{2} \cdot S_r}{\sqrt{3} \cdot U_{tap,s}} \qquad U_{base,s} = \frac{\sqrt{2} \cdot U_{tap,s}}{\sqrt{3}} \tag{39}$$

$$\Psi_{base} = \frac{U_{base,p}}{\omega_N} \tag{40}$$

where:

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

S_r is the rated apparent power S_r in [VA] (*t:Snom*)

ω_N is the nominal angular velocity in [rad/s], where:

$$\omega_N = 2 \cdot \pi \cdot f_N$$

f_N is the nominal frequency of the network in [Hz]

***Wye*-connected voltage transformer:**

Secondary voltages:

$$U_{s,A} = u_{e,A} \cdot U_{base,s} \tag{41}$$
$$U_{s,B} = u_{e,A} \cdot U_{base,s} \tag{42}$$
$$U_{s,C} = u_{e,A} \cdot U_{base,s} \tag{43}$$

Voltage equations:

$$u_{p,A} = (r_t \cdot i_{leak,A} + \frac{d}{dt} i_{leak,A} \cdot l_t) + (r_b \cdot i_{b,A} + \frac{d}{dt} i_{b,A} \cdot l_b) = u_{leak,A} + u_{e,A} \tag{44}$$
$$u_{p,B} = (r_t \cdot i_{leak,B} + \frac{d}{dt} i_{leak,B} \cdot l_t) + (r_b \cdot i_{b,B} + \frac{d}{dt} i_{b,B} \cdot l_b) = u_{leak,B} + u_{e,B} \tag{45}$$
$$u_{p,C} = (r_t \cdot i_{leak,C} + \frac{d}{dt} i_{leak,C} \cdot l_t) + (r_b \cdot i_{b,C} + \frac{d}{dt} i_{b,C} \cdot l_b) = u_{leak,C} + u_{e,C} \tag{46}$$

Magnetic flux equations:

$$\frac{d}{dt}\psi_A = \omega_N \cdot u_{e,A} \tag{47}$$
$$\frac{d}{dt}\psi_B = \omega_N \cdot u_{e,B} \tag{48}$$
$$\frac{d}{dt}\psi_C = \omega_N \cdot u_{e,C} \tag{49}$$

Current equations:

$$i_{leak,A} = u_{e,A} \cdot \frac{1}{r_m} + i_{e,A} + i_{b,A} \tag{50}$$
$$i_{leak,B} = u_{e,B} \cdot \frac{1}{r_m} + i_{e,B} + i_{b,B} \tag{51}$$
$$i_{leak,C} = u_{e,C} \cdot \frac{1}{r_m} + i_{e,C} + i_{b,C} \tag{52}$$
$$i_{p,A} = \frac{d}{dt} u_{p,A} \cdot c_t + i_{leak,A} = i_{stray,A} + i_{leak,A} \tag{53}$$
$$i_{p,B} = \frac{d}{dt} u_{p,B} \cdot c_t + i_{leak,B} = i_{stray,B} + i_{leak,B} \tag{54}$$
$$i_{p,C} = \frac{d}{dt} u_{p,C} \cdot c_t + i_{leak,C} = i_{stray,C} + i_{leak,C} \tag{55}$$
$$\tag{56}$$

Open delta-connected voltage transformer:

Secondary voltages:

$$U_{s,A} = u_{e,A} \cdot U_{base,s} \tag{57}$$
$$U_{s,B} = 0 \tag{58}$$
$$U_{s,C} = u_{e,C} \cdot U_{base,s} \tag{59}$$

Voltage equations:

$$(u_{p,A} - u_{p,B}) = (r_t \cdot i_{leak,A} + \frac{d}{dt} i_{leak,A} \cdot l_t) + (r_b \cdot i_{b,A} + \frac{d}{dt} i_{b,A} \cdot l_b) = u_{leak,A} + u_{e,A} \tag{60}$$
$$0 = (r_t \cdot i_{leak,B} + \frac{d}{dt} i_{leak,B} \cdot l_t) + (r_b \cdot i_{b,B} + \frac{d}{dt} i_{b,B} \cdot l_b) = u_{leak,B} + u_{e,B} \tag{61}$$
$$(u_{p,C} - u_{p,B}) = (r_t \cdot i_{leak,C} + \frac{d}{dt} i_{leak,C} \cdot l_t) + (r_b \cdot i_{b,C} + \frac{d}{dt} i_{b,C} \cdot l_b) = u_{leak,C} + u_{e,C} \tag{62}$$

Magnetic flux equations:

$$\frac{d}{dt}\psi_A = \omega_N \cdot u_{e,A} \tag{63}$$

$$\frac{d}{dt}\psi_B = 0 \tag{64}$$

$$\frac{d}{dt}\psi_C = \omega_N \cdot u_{e,C} \tag{65}$$

Current equations:

$$i_{leak,A} = u_{e,A} \cdot \frac{1}{r_m} + i_{e,A} + i_{b,A} \tag{66}$$

$$i_{leak,B} = 0 \tag{67}$$

$$i_{leak,C} = u_{e,C} \cdot \frac{1}{r_m} + i_{e,C} + i_{b,C} \tag{68}$$

$$i_{p,A} = \frac{d}{dt}(u_{p,A} - u_{p,B}) \cdot c_t + i_{leak,A} = i_{stray,A} + i_{leak,A} \tag{69}$$

$$i_{p,B} = 0 \tag{70}$$

$$i_{p,C} = \frac{d}{dt}(u_{p,C} - u_{p,B}) \cdot c_t + i_{leak,C} = i_{stray,C} + i_{leak,C} \tag{71}$$

where:

Note: *Per-unit* quantities are referred to the primary-side base system, unless stated otherwise.

$U_{s,X}$ is the secondary voltage, phase X in [V] (*s:U2r_X*)

$u_{p,X}$ is the primary voltage, phase X in [p.u.] (see also Section 2.1)

$u_{e,X}$ is the excitation voltage, phase X in [p.u.]

$i_{e,X}$ is the excitation current, phase X in [p.u.] (*c:ie_X*; see also Section 2.6)

$i_{leak,X}$ is the leakage current, phase X in [p.u.] (*s:ileak_X*)

$i_{b,X}$ is the burden current, phase X in [p.u.] (*s:ib_X*)

$i_{stray,X}$ is the current into the stray capacitance, phase X in [p.u.] (*c:istray_X*)

$i_{p,X}$ is the current into the whole voltage transformer, phase X in [p.u.] (*s:ip_X*)

ψ_X is the magnetic flux, phase X in [p.u.], referred to Ψ_{base} (*s:psim_X*)

r_t is the leakage resistance in [p.u.] (*t:Rt*)

l_t is the leakage inductance in [p.u.] (*t:Lt*)

c_t is the stray capacitance in [p.u.] (*e:Ct*)

r_b is the actual burden resistance in [p.u.], referred to $Z_{base,s}$ (*e:Rb*)

l_b is the actual burden inductance in [p.u.], referred to $Z_{base,s}$ (*e:Lb*)

ω_N is the nominal angular velocity in [rad/s], where:

$$\omega_N = 2 \cdot \pi \cdot f_N$$

f_N is the nominal frequency of the network in [Hz]

For 2-phase, *wye*-connected voltage transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Table 2.7: Signals, detailed 2- or 3-phase VT, EMT simulation

Signal	Description	Unit	Type
U_A	Primary Voltage A	p.u.	IN
U_B	Primary Voltage B	p.u.	IN
U_C	Primary Voltage C	p.u.	IN
U2r_A	Secondary Voltage A	V	OUT
U2r_B	Secondary Voltage B	V	OUT
U2r_C	Secondary Voltage C	V	OUT
U2r_A:dt	Secondary Voltage, derivative A	V/s	OUT
U2r_B:dt	Secondary Voltage, derivative B	V/s	OUT
U2r_C:dt	Secondary Voltage, derivative C	V/s	OUT
ileak_A	Leakage Current A	p.u.	STATE
ileak_B	Leakage Current B	p.u.	STATE
ileak_C	Leakage Current C	p.u.	STATE
ib_A	Burden Current A	p.u.	STATE
ib_B	Burden Current B	p.u.	STATE
ib_C	Burden Current C	p.u.	STATE
ip_A	Primary Current A	p.u.	STATE
ip_B	Primary Current B	p.u.	STATE
ip_C	Primary Current C	p.u.	STATE
psim_A	Mag. Flux A	p.u.	STATE
psim_B	Mag. Flux B	p.u.	STATE
psim_C	Mag. Flux C	p.u.	STATE
ileak_A:dt	Leakage Current, derivative A	p.u./s	d/dt
ileak_B:dt	Leakage Current, derivative B	p.u./s	d/dt
ileak_C:dt	Leakage Current, derivative C	p.u./s	d/dt
ib_A:dt	Burden Current, derivative A	p.u./s	d/dt
ib_B:dt	Burden Current, derivative B	p.u./s	d/dt
ib_C:dt	Burden Current, derivative C	p.u./s	d/dt
ip_A:dt	Primary Current, derivative A	p.u./s	d/dt
ip_B:dt	Primary Current, derivative B	p.u./s	d/dt
ip_C:dt	Primary Current, derivative C	p.u./s	d/dt
psim_A:dt	Mag. Flux, derivative A	p.u./s	d/dt
psim_B:dt	Mag. Flux, derivative B	p.u./s	d/dt
psim_C:dt	Mag. Flux, derivative C	p.u./s	d/dt

Table 2.8: Calculation parameters, detailed 2- or 3-phase VT, EMT simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Usec:A	Secondary voltage A	V
Usec:B	Secondary voltage B	V
Usec:C	Secondary voltage C	V
ie_A	Excitation Current A	p.u.
ie_B	Excitation Current B	p.u.
ie_C	Excitation Current C	p.u.
istray_A	Stray Current A	p.u.
istray_B	Stray Current B	p.u.
istray_C	Stray Current C	p.u.

The calculation parameters *c:Usec:A*, *c:Usec:B* and *c:Usec:C* show the magnitudes of the corresponding secondary voltages.

2.4 Capacitive 2- or 3-Phase Voltage Transformer

The capacitive voltage transformer combines a capacitive voltage divider with a detailed inductive voltage transformer. It also includes compensation circuits commonly found in CVTs. The model is based on [3], [2] and [1].

Note: Capacitive voltage transformers can only be modelled as *wye*-connected transformers.

2.4.1 Load Flow, Short-Circuit Calculation and RMS Simulation

Saturation is considered with a steady state equivalent. The equivalent circuit is shown in Figure 2.3.

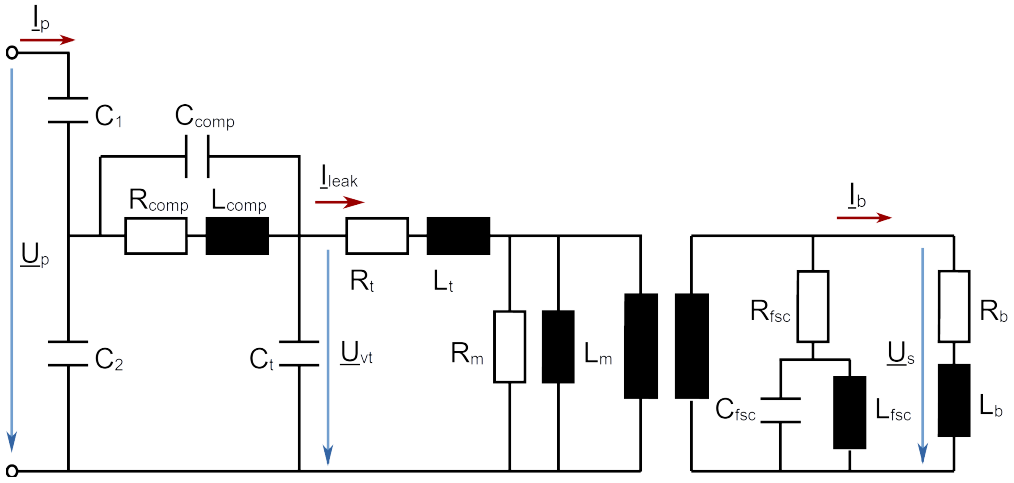


Figure 2.3: Equivalent circuit for a capacitive VT used in load flow, short-circuit calculation and RMS simulation

Some of the equations below are given in *per-unit* for readability. The relevant system bases are calculated as follows:

$$Z_{base,p} = \frac{U_{tap,p}^2}{S_r} \qquad I_{base,p} = \frac{S_r}{\sqrt{3} \cdot U_{tap,p}} \qquad U_{base,p} = \frac{U_{tap,p}}{\sqrt{3}} \tag{72}$$

$$Z_{base,i} = \frac{U_i^2}{S_r} \qquad I_{base,i} = \frac{S_r}{\sqrt{3} \cdot U_i} \qquad U_{base,i} = \frac{U_i}{\sqrt{3}} \tag{73}$$

$$Z_{base,s} = \frac{U_{tap,s}^2}{S_r} \qquad I_{base,s} = \frac{S_r}{\sqrt{3} \cdot U_{tap,s}} \qquad U_{base,s} = \frac{U_{tap,s}}{\sqrt{3}} \tag{74}$$

where:

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

U_i is the nominal intermediate voltage in [V] (*t:Uinom*)

S_r is the rated apparent power S_r in [VA] (*t:Snom*)

Various equivalent impedances, derived from Figure 2.3 and used in the equations for output signals and calculation parameters, are calculated as follows:

Complex impedances:

$$\underline{z}_b = \cdot(R_b + j \cdot X_b) \cdot \frac{1}{Z_{base,s}} \tag{75}$$

$$\underline{z}_{fsc} = \left(R_{fsc} + j \cdot \frac{X_{L,fsc} \cdot X_{C,fsc}}{X_{L,fsc} + X_{C,fsc}}\right) \cdot \frac{1}{Z_{base,s}} \tag{76}$$

$$\underline{z}_m = \frac{R_m \cdot j \cdot X_m}{R_m + j \cdot X_m} \cdot \frac{1}{Z_{base,i}} \tag{77}$$

$$\underline{z}_t = (R_t + j \cdot X_t) \cdot \frac{1}{Z_{base,i}} \tag{78}$$

$$\underline{z}_{C,t} = j \cdot X_{C,t} \cdot \frac{1}{Z_{base,i}} \tag{79}$$

$$\underline{z}_{comp} = \frac{(R_{comp} + j \cdot X_{L,comp}) \cdot j \cdot X_{C,comp}}{(R_{comp} + j \cdot X_{L,comp}) + j \cdot X_{C,comp}} \cdot \frac{1}{Z_{base,i}} \tag{80}$$

$$\underline{z}_{C1} = j \cdot X_{C1} \cdot \frac{1}{Z_{base,i}} \tag{81}$$

$$\underline{z}_{C2} = j \cdot X_{C2} \cdot \frac{1}{Z_{base,i}} \tag{82}$$

Equivalent impedances:

$$\underline{z}_{sec} = \frac{\underline{z}_{fsc} \cdot \underline{z}_b}{\underline{z}_{fsc} + \underline{z}_b} \tag{83}$$

$$\underline{z}_{m\|sec} = \frac{\underline{z}_m \cdot \underline{z}_{sec}}{\underline{z}_m + \underline{z}_{sec}} \tag{84}$$

$$\underline{k}_{ind} = \frac{\underline{z}_{m\|sec}}{\underline{z}_t + \underline{z}_{m\|sec}} \tag{85}$$

$$\underline{z}_{vt} = \frac{\underline{z}_{C,t} \cdot (\underline{z}_t + \underline{z}_{m\|sec})}{\underline{z}_{C,t} + (\underline{z}_t + \underline{z}_{m\|sec})} \tag{86}$$

$$\underline{z}_i = \underline{z}_{comp} + \underline{z}_{vt} \tag{87}$$

$$\underline{z}_e = \underline{z}_{C1} + \frac{\underline{z}_{C2} \cdot \underline{z}_i}{\underline{z}_{C2} + \underline{z}_i} \tag{88}$$

where:

R_b is the actual burden resistance in [Ω] (*e:Rb*)

X_b is the actual burden reactance in [Ω], where:

$$X_b = \omega_N \cdot L_b$$

L_b is the actual burden inductance in [H] (*e:Lb*)

R_{fsc} is the ferroresonance suppression coil resistance in [Ω] (*t:Rfsc*)

$X_{L,fsc}$ is the ferroresonance suppression coil inductive reactance in [Ω], where:

$$X_{L,fsc} = \omega_N \cdot L_{fsc}$$

L_{fsc} is the ferroresonance suppression coil inductance in [H] (*t:Lfsc*)

$X_{C,fsc}$ is the ferroresonance suppression coil capacitive reactance in [Ω], where:

$$X_{C,fsc} = -\frac{1}{\omega_N \cdot C_{fsc} \cdot 10^{-9}}$$

C_{fsc} is the ferroresonance suppression coil capacitance in [nF] (*t:Cfsc*)

R_m is the magnetising resistance in [Ω] (*t:Rm*)

X_m is the magnetising reactance in [Ω], where:

$$X_m = \omega_N \cdot L_m$$

L_b is the magnetising inductance in [H] (*t:Lm*)

R_t is the leakage resistance in [Ω] (*t:Rt*)

X_t is the leakage reactance in [Ω], where:

$$X_t = \omega_N \cdot L_t$$

L_t is the leakage inductance in [H] (*t:Lt*)

$X_{C,t}$ is the stray capacitance reactance in [Ω], where:

$$X_{C,fsc} = -\frac{1}{\omega_N \cdot C_t \cdot 10^{-9}}$$

C_t is the stray capacitance in [nF] (*e:Ct*)

R_{comp} is the compensation coil resistance in [Ω] (*t:Rcomp*)

$X_{L,comp}$ is the compensation coil inductive reactance in [Ω], where:

$$X_{L,comp} = \omega_N \cdot L_{comp}$$

L_{comp} is the suppression coil inductance in [H] (*t:Lcomp*)

$X_{C,comp}$ is the compensation coil capacitive reactance in [Ω], where:

$$X_{C,comp} = -\frac{1}{\omega_N \cdot C_{comp} \cdot 10^{-9}}$$

C_{comp} is the suppression coil capacitance in [nF] (*e:Ccomp*)

X_{C1} is the primary capacitance reactance in [Ω], where:

$$X_{C1} = -\frac{1}{\omega_N \cdot C_1 \cdot 10^{-9}}$$

C_1 is the primary capacitance in [nF] (*t:C1*)

X_{C2} is the secondary capacitance reactance in [Ω], where:

$$X_{C1} = -\frac{1}{\omega_N \cdot C_2 \cdot 10^{-9}}$$

C_2 is the secondary capacitance in [nF] (*t:C2*)

$\underline{k}_{ind}$ is the complex factor for the inductive voltage divider

ω_N is the nominal angular velocity in [rad/s], where:

$$\omega_N = 2 \cdot \pi \cdot f_N$$

f_N is the nominal frequency of the network in [Hz]

Secondary voltages:

$$\underline{U}_{s,A} = \underline{u}_{b,A} \cdot U_{base,s} \quad (89)$$
$$\underline{U}_{s,B} = \underline{u}_{b,B} \cdot U_{base,s} \quad (90)$$
$$\underline{U}_{s,C} = \underline{u}_{b,C} \cdot U_{base,s} \quad (91)$$

Intermediate voltages:

$$\underline{u}_{b,A} = \underline{u}_{vt,A} \cdot \underline{k}_{ind} \quad (92)$$
$$\underline{u}_{b,B} = \underline{u}_{vt,B} \cdot \underline{k}_{ind} \quad (93)$$
$$\underline{u}_{b,C} = \underline{u}_{vt,C} \cdot \underline{k}_{ind} \quad (94)$$
$$\underline{u}_{vt,A} = \underline{u}_{i,A} - \underline{i}_{comp,A} \cdot \underline{z}_{comp} \quad (95)$$
$$\underline{u}_{vt,B} = \underline{u}_{i,B} - \underline{i}_{comp,B} \cdot \underline{z}_{comp} \quad (96)$$
$$\underline{u}_{vt,C} = \underline{u}_{i,C} - \underline{i}_{comp,C} \cdot \underline{z}_{comp} \quad (97)$$
$$\underline{u}_{i,A} = \underline{u}_{p,A} - \underline{i}_{p,A} \cdot \underline{z}_{C1} \quad (98)$$
$$\underline{u}_{i,A} = \underline{u}_{p,B} - \underline{i}_{p,B} \cdot \underline{z}_{C1} \quad (99)$$
$$\underline{u}_{i,A} = \underline{u}_{p,C} - \underline{i}_{p,C} \cdot \underline{z}_{C1} \quad (100)$$

Intermediate currents:

$$\underline{i}_{p,A} = \frac{\underline{u}_{p,A}}{\underline{z}_e} \quad (101)$$
$$\underline{i}_{p,B} = \frac{\underline{u}_{p,B}}{\underline{z}_e} \quad (102)$$
$$\underline{i}_{p,C} = \frac{\underline{u}_{p,C}}{\underline{z}_e} \quad (103)$$
$$\underline{i}_{comp,A} = \frac{\underline{u}_{i,A}}{\underline{z}_i} \quad (104)$$
$$\underline{i}_{comp,B} = \frac{\underline{u}_{i,B}}{\underline{z}_i} \quad (105)$$
$$\underline{i}_{comp,C} = \frac{\underline{u}_{i,C}}{\underline{z}_i} \quad (106)$$

where:

Note: *Per-unit* quantities are referred to the intermediate base system, unless stated otherwise.

$\underline{U}_{s,X}$ is the secondary voltage, phase X in [V] (*s:U2r_X* $+ j \cdot$ *s:U2i_X*)

$\underline{u}_{p,X}$ is the primary voltage, phase X in [p.u.] (see also Section 2.1)

$\underline{u}_{b,X}$ is the burden voltage, phase X in [p.u.]

$\underline{u}_{vt,X}$ is the voltage seen by the inductive transformer, phase X in [p.u.]

$\underline{u}_{i,X}$ is the intermediate circuit voltage, phase X in [p.u.]

$\underline{i}_{p,X}$ is the current flowing into the whole voltage transformer , phase X in [p.u.]

$\underline{i}_{comp,X}$ is current flowing into the compensation branch, phase X in [p.u.]

For 2-phase voltage transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Signal connections are identical to the ideal voltage transformer. See Tables 2.1 and 2.2.

Table 2.9: Calculation parameters, capacitive 2- or 3-phase VT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Usec:A	Secondary voltage A	V
Usec:B	Secondary voltage B	V
Usec:C	Secondary voltage C	V
ip_A	Primary Current A	p.u.
ip_B	Primary Current B	p.u.
ip_C	Primary Current C	p.u.
ib_A	Burden Current A	p.u.
ib_B	Burden Current B	p.u.
ib_C	Burden Current C	p.u.
ileak_A	Leakage Current A	p.u.
ileak_B	Leakage Current B	p.u.
ileak_C	Leakage Current C	p.u.
ub_A	Burden Voltage A	p.u.
ub_B	Burden Voltage B	p.u.
ub_C	Burden Voltage C	p.u.
Zb:r	Burden, Impedance, Real Part	Ω
Zb:i	Burden, Impedance, Imaginary Part	Ω
Zb:phi	Burden, Impedance, Angle	$^\circ$
Zb:mag	Burden, Impedance, Magnitude	Ω
Zb:phirad	Burden, Impedance, Angle	rad
ui_A	Intermediate Circuit Voltage A	p.u.
ui_B	Intermediate Circuit Voltage B	p.u.
ui_C	Intermediate Circuit Voltage C	p.u.
icomp_A	Compensation Current A	p.u.
icomp_B	Compensation Current B	p.u.
icomp_C	Compensation Current C	p.u.
irlcomp_A	Compensation Current, RL-Branch A	p.u.
irlcomp_B	Compensation Current, RL-Branch B	p.u.
irlcomp_C	Compensation Current, RL-Branch C	p.u.
uvt_A	Intermediate VT Voltage A	p.u.
uvt_B	Intermediate VT Voltage B	p.u.
uvt_C	Intermediate VT Voltage C	p.u.
ifsc_A	Ferroresonance Current A	p.u.
ifsc_B	Ferroresonance Current B	p.u.
ifsc_C	Ferroresonance Current C	p.u.

The calculation parameters *c:Usec:A*, *c:Usec:B* and *c:Usec:C* show the magnitudes of the corresponding complex secondary voltages.

Most of the calculation quantities correspond to the magnitudes of the corresponding complex value used in the equations. Quantities which are not explicitly calculated in the equations are explained below.

Note: *Per-unit* quantities are referred to the intermediate base system

The calculation parameters *c:ib_A*, *c:ib_B* and *c:ib_C* show the complex current magnitudes flowing into the burden and are calculated as follows:

$$i_{b,X} = \left| \frac{\underline{u}_{b,X}}{\underline{z}_b} \right| \tag{107}$$

The calculation parameters *c:ileak_A*, *c:ileak_B* and *c:ileak_C* show the complex current magnitudes flowing into the leakage branch of the inductive transformer and are calculated as follows:

$$i_{leak,X} = \left| \frac{\underline{u}_{vt,X}}{\underline{z}_t + \underline{z}_{m\|sec}} \right| \tag{108}$$

The calculation parameters *c:irlcomp_A*, *c:irlcomp_B* and *c:irlcomp_C* show the complex current magnitudes flowing into the RL-branch of the compensation and are calculated as follows:

$$i_{rlcomp,X} = \left| \frac{\underline{u}_{i,X} - \underline{u}_{vt,X}}{r_{comp} + j \cdot x_{L,comp}} \right| \tag{109}$$

The calculation parameters *c:ifsc_A*, *c:ifsc_B* and *c:ifsc_C* show the complex current magnitudes flowing into the ferroresonance suppresion coil and are calculated as follows:

$$i_{b,X} = \left| \frac{\underline{u}_{b,X}}{\underline{z}_{fsc}} \right| \tag{110}$$

2.4.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation. The saturation is considered with a non-linear magnetising inductance. The equivalent circuit is shown in Figure 2.4.

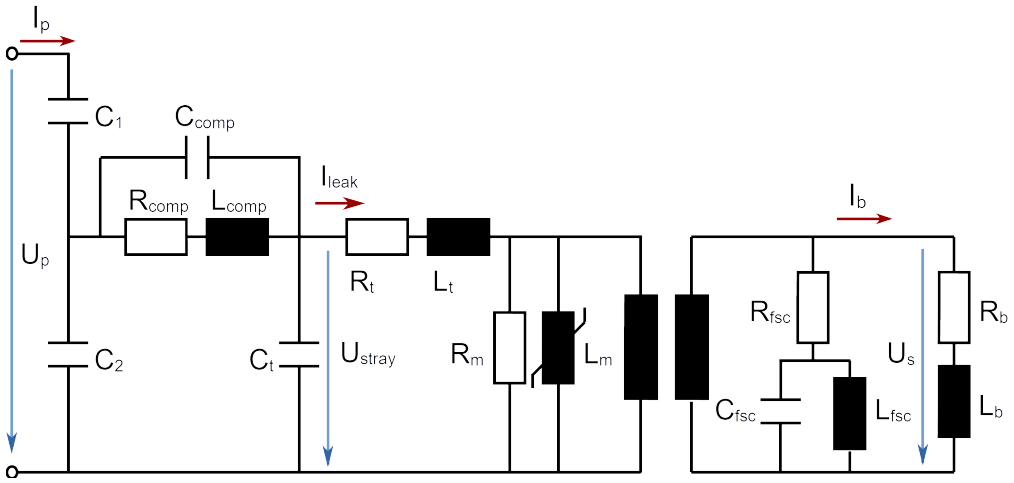


Figure 2.4: Equivalent circuit for a capacitive VT used in EMT simulation

$$Z_{base,p} = \frac{U_{tap,p}^2}{S_r} \qquad I_{base,p} = \frac{\sqrt{2} \cdot S_r}{\sqrt{3} \cdot U_{tap,p}} \qquad U_{base,p} = \frac{\sqrt{2} \cdot U_{tap,p}}{\sqrt{3}} \tag{111}$$

$$Z_{base,i} = \frac{U_i^2}{S_r} \qquad I_{base,i} = \frac{\sqrt{2} \cdot S_r}{\sqrt{3} \cdot U_i} \qquad U_{base,i} = \frac{\sqrt{2} \cdot U_i}{\sqrt{3}} \tag{112}$$

$$Z_{base,s} = \frac{U_{tap,s}^2}{S_r} \qquad I_{base,s} = \frac{\sqrt{2} \cdot S_r}{\sqrt{3} \cdot U_{tap,s}} \qquad U_{base,s} = \frac{\sqrt{2} \cdot U_{tap,s}}{\sqrt{3}} \tag{113}$$

$$\Psi_{base} = \frac{U_{base,p}}{\omega_N} \tag{114}$$

where:

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

U_i is the nominal intermediate voltage in [V] (*t:Uinom*)

S_r is the rated apparent power S_r in [VA] (*t:Snom*)

ω_N is the nominal angular velocity in [rad/s], where:

$$\omega_N = 2 \cdot \pi \cdot f_N$$

f_N is the nominal frequency of the network in [Hz]

Secondary voltages:

$$U_{s,A} = u_{e,A} \cdot U_{base,s} \tag{115}$$

$$U_{s,B} = u_{e,A} \cdot U_{base,s} \tag{116}$$

$$U_{s,C} = u_{e,A} \cdot U_{base,s} \tag{117}$$

Voltage equations:

$$u_{e,A} = u_{p,A} - u_{comp,A} - u_{leak,A} = u_{b,A} = u_{fsc,A} \quad (118)$$

$$u_{e,B} = u_{p,B} - u_{comp,B} - u_{leak,B} = u_{b,B} = u_{fsc,B} \quad (119)$$

$$u_{e,C} = u_{p,C} - u_{comp,C} - u_{leak,C} = u_{b,C} = u_{fsc,C} \quad (120)$$

$$u_{b,A} = r_b \cdot i_{b,A} + \frac{d}{dt} i_{b,A} \cdot l_b \quad (121)$$

$$u_{b,B} = r_b \cdot i_{b,B} + \frac{d}{dt} i_{b,B} \cdot l_b \quad (122)$$

$$u_{b,C} = r_b \cdot i_{b,C} + \frac{d}{dt} i_{b,C} \cdot l_b \quad (123)$$

$$u_{fsc,A} = r_{fsc} \cdot i_{fsc,A} + \frac{d}{dt} i_{L,fsc,A} \cdot l_{fsc} = r_{fsc} \cdot i_{fsc,A} + u_{C,fsc,A} \quad (124)$$

$$u_{fsc,B} = r_{fsc} \cdot i_{fsc,B} + \frac{d}{dt} i_{L,fsc,B} \cdot l_{fsc} = r_{fsc} \cdot i_{fsc,B} + u_{C,fsc,B} \quad (125)$$

$$u_{fsc,C} = r_{fsc} \cdot i_{fsc,C} + \frac{d}{dt} i_{L,fsc,C} \cdot l_{fsc} = r_{fsc} \cdot i_{fsc,C} + u_{C,fsc,C} \quad (126)$$

$$u_{p,A} = u_{C1,A} + u_{C2,A} = u_{C1,A} + u_{comp,A} + u_{stray,A} \quad (127)$$

$$u_{p,B} = u_{C1,B} + u_{C2,B} = u_{C1,B} + u_{comp,B} + u_{stray,B} \quad (128)$$

$$u_{p,C} = u_{C1,C} + u_{C2,C} = u_{C1,C} + u_{comp,C} + u_{stray,C} \quad (129)$$

$$u_{comp,A} = r_{comp} \cdot i_{rlcomp,A} + \frac{d}{dt} i_{RL,comp,A} \cdot l_{comp} \quad (130)$$

$$u_{comp,B} = r_{comp} \cdot i_{rlcomp,B} + \frac{d}{dt} i_{RL,comp,B} \cdot l_{comp} \quad (131)$$

$$u_{comp,C} = r_{comp} \cdot i_{rlcomp,C} + \frac{d}{dt} i_{RL,comp,C} \cdot l_{comp} \quad (132)$$

$$u_{leak,A} = r_t \cdot i_{leak,A} + \frac{d}{dt} i_{leak,A} \cdot l_t \quad (133)$$

$$u_{leak,B} = r_t \cdot i_{leak,B} + \frac{d}{dt} i_{leak,B} \cdot l_t \quad (134)$$

$$u_{leak,C} = r_t \cdot i_{leak,C} + \frac{d}{dt} i_{leak,C} \cdot l_t \quad (135)$$

Magnetic flux equations:

$$\frac{d}{dt} \psi_A = \omega_N \cdot u_{e,A} \quad (136)$$

$$\frac{d}{dt} \psi_B = \omega_N \cdot u_{e,B} \quad (137)$$

$$\frac{d}{dt} \psi_C = \omega_N \cdot u_{e,C} \quad (138)$$

Current equations:

$$i_{leak,A} = u_{e,A} \cdot \frac{1}{r_m} + i_{e,A} + i_{fsc,A} + i_{b,A} \quad (139)$$

$$i_{leak,B} = u_{e,B} \cdot \frac{1}{r_m} + i_{e,B} + i_{fsc,B} + i_{b,B} \quad (140)$$

$$i_{leak,C} = u_{e,C} \cdot \frac{1}{r_m} + i_{e,C} + i_{fsc,C} + i_{b,C} \quad (141)$$

$$i_{fsc,A} = \frac{u_{e,A} - u_{C,fsc,A}}{r_{fsc}} = \frac{d}{dt} u_{C,fsc,A} \cdot c_{fsc} + i_{L,fsc,A} \quad (142)$$

$$i_{fsc,B} = \frac{u_{e,B} - u_{C,fsc,B}}{r_{fsc}} = \frac{d}{dt} u_{C,fsc,B} \cdot c_{fsc} + i_{L,fsc,B} \quad (143)$$

$$i_{fsc,C} = \frac{u_{e,C} - u_{C,fsc,C}}{r_{fsc}} = \frac{d}{dt} u_{C,fsc,C} \cdot c_{fsc} + i_{L,fsc,C} \quad (144)$$

$$i_{comp,A} = \frac{d}{dt} u_{stray,A} \cdot c_t + i_{leak,A} = \frac{d}{dt} u_{comp,A} \cdot c_{comp} + i_{RL,comp,A} \quad (145)$$

$$i_{comp,B} = \frac{d}{dt} u_{stray,B} \cdot c_t + i_{leak,B} = \frac{d}{dt} u_{comp,B} \cdot c_{comp} + i_{RL,comp,B} \quad (146)$$

$$i_{comp,C} = \frac{d}{dt} u_{stray,C} \cdot c_t + i_{leak,C} = \frac{d}{dt} u_{comp,C} \cdot c_{comp} + i_{RL,comp,C} \quad (147)$$

$$i_{stray,A} = \frac{d}{dt} u_{stray,A} \cdot c_t \quad (148)$$

$$i_{stray,B} = \frac{d}{dt} u_{stray,B} \cdot c_t \quad (149)$$

$$i_{stray,C} = \frac{d}{dt} u_{stray,C} \cdot c_t \quad (150)$$

$$i_{p,A} = \frac{d}{dt} u_{p,A} \cdot c_1 \quad (151)$$

$$i_{p,B} = \frac{d}{dt} u_{p,B} \cdot c_1 \quad (152)$$

$$i_{p,C} = \frac{d}{dt} u_{p,C} \cdot c_1 \quad (153)$$

(154)

where:

Note: *Per-unit* quantities are referred to the intermediate base system, unless stated otherwise.

$U_{s,X}$ is the secondary voltage, phase X in [V] (*s:U2r_X*)

$u_{p,X}$ is the primary voltage, phase X in [p.u.] (see also Section 2.1)

$u_{e,X}$ is the excitation voltage, phase X in [p.u.]

$i_{e,X}$ is the excitation current, phase X in [p.u.] (*c:ie_X*; see also Section 2.6)

$u_{stray,X}$ is the voltage at the stray capacitance, phase X in [p.u.] (*s:ustray_X*)

$i_{leak,X}$ is the leakage current, phase X in [p.u.] (*s:ileak_X*)

$i_{b,X}$ is the burden current, phase X in [p.u.] (*s:ib_X*)

$i_{fsc,X}$ is the ferroresonance suppression current, phase X in [p.u.] (*c:ifsc_X*)

$i_{L,fsc,X}$ is the current following into the *FSC* inductance, phase X in [p.u.]

$i_{stray,X}$ is the current into the stray capacitance, phase X in [p.u.] (*c:istray_X*)

$i_{comp,X}$ is the current into compensation coil, phase X in [p.u.] (*c:icomp_X*)

$i_{p,X}$ is the current into the whole voltage transformer, phase X in [p.u.] (*s:i_X*)

ψ_X is the magnetic flux, phase X in [p.u.], referred to Ψ_{base} (*s:psim_X*)

c_1 is the primary capacitance in [p.u.] (*t:C1*)

c_2 is the secondary capacitance in [p.u.] (*t:C2*)

r_{comp} is the compensation coil resistance in [p.u.] (*t:Rcomp*)

l_{comp} is the compensation coil coil inductance in [p.u.] (*t:Lcomp*)

c_{comp} is the compensation coil capacitance in [p.u.] (*e:Ccomp*)

r_t is the leakage resistance in [p.u.] (*t:Rt*)

l_t is the leakage inductance in [p.u.] (*t:Lt*)

c_t is the stray capacitance in [p.u.] (*e:Ct*)

r_{fsc} is the ferroresonance suppression coil resistance in [p.u.], referred to $Z_{base,s}$ (*t:Rfsc*)

l_{fsc} is the ferroresonance suppression coil inductance in [p.u.], referred to $Z_{base,s}$ (*t:Lfsc*)

c_{fsc} is the ferroresonance suppression coil capacitance in [p.u.], referred to $Z_{base,s}$ (*t:Cfsc*)

r_b is the actual burden resistance in [p.u.], referred to $Z_{base,s}$ (*e:Rb*)

l_b is the actual burden inductance in [p.u.], referred to $Z_{base,s}$ (*e:Lb*)

ω_N is the nominal angular velocity in [rad/s], where:

$$\omega_N = 2 \cdot \pi \cdot f_N$$

f_N is the nominal frequency of the network in [Hz]

For 2-phase voltage transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Table 2.10: Signals, capacitive 2- or 3-phase VT, EMT simulation

Signal	Description	Unit	Type
U_A	Primary Voltage A	p.u.	IN
U_B	Primary Voltage B	p.u.	IN
U_C	Primary Voltage C	p.u.	IN
U2r_A	Secondary Voltage A	V	OUT
U2r_B	Secondary Voltage B	V	OUT
U2r_C	Secondary Voltage C	V	OUT
U2r_A:dt	Secondary Voltage, derivative A	V/s	OUT
U2r_B:dt	Secondary Voltage, derivative B	V/s	OUT
U2r_C:dt	Secondary Voltage, derivative C	V/s	OUT
ileak_A	Leakage Current A	p.u.	STATE
ileak_B	Leakage Current B	p.u.	STATE
ileak_C	Leakage Current C	p.u.	STATE
ib_A	Burden Current A	p.u.	STATE
ib_B	Burden Current B	p.u.	STATE
ib_C	Burden Current C	p.u.	STATE
psim_A	Mag. Flux A	p.u.	STATE
psim_B	Mag. Flux B	p.u.	STATE
psim_C	Mag. Flux C	p.u.	STATE
ustray_A	Intermediate VT Voltage	p.u.	STATE
ustray_B	Intermediate VT Voltage	p.u.	STATE
ustray_C	Intermediate VT Voltage	p.u.	STATE
ileak_A:dt	Leakage Current, derivative A	p.u./s	d/dt
ileak_B:dt	Leakage Current, derivative B	p.u./s	d/dt
ileak_C:dt	Leakage Current, derivative C	p.u./s	d/dt
ib_A:dt	Burden Current, derivative A	p.u./s	d/dt
ib_B:dt	Burden Current, derivative B	p.u./s	d/dt
ib_C:dt	Burden Current, derivative C	p.u./s	d/dt
psim_A:dt	Mag. Flux, derivative A	p.u./s	d/dt
psim_B:dt	Mag. Flux, derivative B	p.u./s	d/dt
psim_C:dt	Mag. Flux, derivative C	p.u./s	d/dt
ustray_A:dt	Intermediate VT Voltage	p.u./s	d/dt
ustray_B:dt	Intermediate VT Voltage	p.u./s	d/dt
ustray_C:dt	Intermediate VT Voltage	p.u./s	d/dt

Table 2.11: Calculation parameters, detailed 2- or 3-phase VT, EMT simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Usec:A	Secondary voltage A	V
Usec:B	Secondary voltage B	V
Usec:C	Secondary voltage C	V
ie_A	Excitation Current A	p.u.
ie_B	Excitation Current B	p.u.
ie_C	Excitation Current C	p.u.
istray_A	Stray Current A	p.u.
istray_B	Stray Current B	p.u.
istray_C	Stray Current C	p.u.
i_A	Primary Current A	p.u.
i_B	Primary Current B	p.u.
i_C	Primary Current C	p.u.
icomp_A	Compensation Current A	p.u.
icomp_B	Compensation Current B	p.u.
icomp_C	Compensation Current C	p.u.
ifsc_A	FSC Current A	p.u.
ifsc_B	FSC Current B	p.u.
ifsc_C	FSC Current C	p.u.

2.5 Ideal Single-Phase Voltage Transformer

For compatibility reasons, single-phase voltage transformers also provide a set of output signals with phase designation (*s:U2r_A* and *s:U2i_A*). These identical to the default signal (*s:U2rs:U2i*) and are therefore not listed in the equations below.

2.5.1 Load Flow, Short-Circuit Calculation and RMS Simulation

Connected to *A*, *B* or *C*:

$$\underline{U}_s = \frac{1}{ratio} \cdot \underline{U}_{p,X} \tag{155}$$

$$3\underline{U}_{s,0} = 0 \tag{156}$$

Connected to *N*:

$$\underline{U}_s = \frac{1}{ratio} \cdot (\underline{U}_{p,A} + \underline{U}_{p,B} + \underline{U}_{p,C}) \tag{157}$$

$$3\underline{U}_{s,0} = \underline{U}_s \tag{158}$$

where:

$\underline{U}_{p,X}$ is the primary voltage, phase X in [V] (see Section 2.1)

$\underline{U}_s$ is the secondary voltage in [V] (*s:U2r* $+ j \cdot$ *s:U2i*)

$3\underline{U}_{s,0}$ is the secondary zero sequence voltage in [V] (*s:U0x3r* $+ j \cdot$ *s:U0x3i*)

$ratio$ is the transmission ratio $ratio = U_{tap,p}/U_{tap,s}$ (*c:ratio*), where:

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

Table 2.12: Signals, single-phase VT, balanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ur_p	Primary Voltage, Real Part	p.u.	IN
Ui_p	Primary Voltage, Imaginary Part	p.u.	IN
U2r	Secondary Voltage, Real Part	V	OUT
U2i	Secondary Voltage, Imaginary Part	V	OUT
U2r_A	Secondary Voltage, Real Part	V	OUT
U2i_A	Secondary Voltage, Imaginary Part	V	OUT
U0x3r	3*U0, Real Part	V	OUT
U0x3i	3*U0, Imaginary Part	V	OUT

Table 2.13: Signals, ideal single-phase VT, unbalanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ur_A	Primary Voltage, Real Part A	p.u.	IN
Ui_A	Primary Voltage, Imaginary Part A	p.u.	IN
Ur_B	Primary Voltage, Real Part B	p.u.	IN
Ui_B	Primary Voltage, Imaginary Part B	p.u.	IN
Ur_C	Primary Voltage, Real Part C	p.u.	IN
Ui_C	Primary Voltage, Imaginary Part C	p.u.	IN
U2r	Secondary Voltage, Real Part	V	OUT
U2i	Secondary Voltage, Imaginary Part	V	OUT
U2r_A	Secondary Voltage, Real Part	V	OUT
U2i_A	Secondary Voltage, Imaginary Part	V	OUT
U0x3r	3*U0, Real Part	V	OUT
U0x3i	3*U0, Imaginary Part	V	OUT

Table 2.14: Calculation parameters, ideal single-phase VT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Usec	Secondary voltage	V
U0x3	3*U0 Secondary voltage	V

The calculation parameters *c:Usec* and *c:U0x3* show the magnitudes of the corresponding complex secondary voltages.

2.5.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation.

Connected to *A*, *B* or *C*:

$$U_s = \frac{1}{ratio} \cdot U_{p,X} \tag{159}$$

$$3U_{s,0} = 0 \tag{160}$$

Connected to *N*:

$$U_s = \frac{1}{ratio} \cdot (U_{p,A} + U_{p,B} + U_{p,C}) \tag{161}$$

$$3U_{s,0} = U_s \tag{162}$$

where:

$U_{p,X}$ is the primary voltage, phase X in [V] (see Section 2.1)

U_s is the secondary voltage in [V] (*s:U2r*)

$3U_{s,0}$ is the secondary zero sequence voltage in [V] (*s:U0x3r*)

$ratio$ is the transmission ratio $ratio = U_{tap,p}/U_{tap,s}$ (*c:ratio*), where:

$U_{tap,p}$ is the selected primary voltage in [V] (*e:ptapset*)

$U_{tap,s}$ is the selected secondary voltage in [V] (*e:stapset*)

Table 2.15: Signals, ideal single-phase VT, EMT simulation

Signal	Description	Unit	Type
U_A	Primary Voltage A	p.u.	IN
U_B	Primary Voltage B	p.u.	IN
U_C	Primary Voltage C	p.u.	IN
U2r	Secondary Voltage	V	OUT
U2r_A	Secondary Voltage	V	OUT
U0x3r	3*U0	V	OUT

2.6 Saturation Characteristic

This section describes the saturation characteristics available for EMT simulation.

2.6.1 Piecewise linear saturation

$$I_{e,X} = \begin{cases} I_{knee} + \dfrac{1}{L_{m,sat}} \cdot (\Psi_X - \Psi_{knee}) & \text{if } \Psi_X > \Psi_{knee} \\ \dfrac{1}{L_m} \cdot \Psi_X & \text{if } -\Psi_{knee} \leq \Psi_X \leq \Psi_{knee} \\ -I_{knee} + \dfrac{1}{L_{m,sat}} \cdot (\Psi_X + \Psi_{knee}) & \text{if } \Psi_X < -\Psi_{knee} \end{cases} \quad (163)$$

where:

$I_{e,X}$ is the excitation current, phase X in [A]

Ψ_X is the magnetic flux, phase X in [Vs]

Ψ_{knee} is the knee-point flux in [Vs]

I_{knee} is the knee-point current in [A]

L_m is the unsaturated magnetising inductance in [H] (*t:Lm*)

$L_{m,sat}$ is the saturated magnetising inductance in [H] (*t:Lmsat*)

The relevant parameters at knee-point are calculated as follows:

$$\Psi_{knee} = \psi_{inp} \cdot \Psi_{base} \quad (164)$$

$$I_{knee} = \frac{\Psi_{knee}}{L_m} \quad (165)$$

where:

ψ_{inp} is the knee flux input in [p.u.] (*t:psi0*)

L_m is the unsaturated magnetising inductance in [H] (*t:Lm*)

2.6.2 Polynomial saturation

$$I_{e,X} = \begin{cases} I_{knee} + \dfrac{1}{L_{m,sat}} \cdot (\Psi_X - \Psi_{knee}) & \text{if } \Psi_X > \Psi_{knee} \\ \dfrac{1}{L_m} \cdot \Psi_X \cdot \left(1 + \left|\dfrac{\Psi_X}{\Psi_0}\right|^{k_{sat}}\right) & \text{if } -\Psi_{knee} \leq \Psi_X \leq \Psi_{knee} \\ -I_{knee} + \dfrac{1}{L_{m,sat}} \cdot (\Psi_X + \Psi_{knee}) & \text{if } \Psi_X < -\Psi_{knee} \end{cases} \quad (166)$$

where:

$I_{e,X}$ is the excitation current, phase X in [A]

Ψ_X is the magnetic flux, phase X in [Vs]

Ψ_{knee} is the knee-point flux in [Vs]

Ψ_0 is the nominal flux in [Vs]

I_{knee} is the knee-point current in [A]

k_{sat} is the saturation exponent (*e:ksat*)

L_m is the unsaturated magnetising inductance in [H] (*t:Lm*)

$L_{m,sat}$ is the saturated magnetising inductance in [H] (*t:Lmsat*)

The relevant parameters at knee-point and the nominal flux are calculated as follows:

$$\Psi_{knee} = \psi_{inp} \cdot \Psi_{base} \cdot \frac{k_{sat}+1}{k_{sat}} \tag{167}$$

$$k_0 = \frac{-\ln\left(\frac{\left(\frac{L_m}{L_m,sat}\right)-1}{k_{sat}+1}\right)}{k_{sat}}$$

$$\Psi_0 = \Psi_{knee} \cdot e^{k_0} \tag{168}$$

$$I_{knee} = \frac{\Psi_{knee}}{L_m} \cdot \left(\frac{L_m}{L_m,sat}\right)^{k_{sat}} \tag{169}$$

where:

ψ_{inp} is the knee flux input in [p.u.] (*t:psi0*)

L_m is the unsaturated magnetising inductance in [H] (*t:Lm*)

$L_{m,sat}$ is the saturated magnetising inductance in [H] (*t:Lmsat*)

3 References

[1] D. Fernandes Jr., W. L. A. Neves, and J. C. A. Vasconcelos. Identification of parameters for coupling capacitor voltage transformers. *Proceedings of IPST*, (01IPST080), 2001.

[2] M. Kezunovic, L. Kojovic, V. Skendzic, C. W. Fromen, D. R. Sevcik, and S. L. Nilsson. Digital models of coupling capacitor voltage transformers for protective relay transient studies. *IEEE Transactions on Power Delivery*, 7(4), 1992.

[3] L. Kojovic, M. Kezunovic, and C.W. Fromen. A new method for the ccvt performance analysis using field measurements, signal processing and emtp modeling. *IEEE Transactions on Power Delivery*, 9(4), 1994.

List of Figures

1.1 3-phase voltage transformer connections 3

1.2 1-phase voltage transformer in broken delta connection (N) 3

2.1 Equivalent circuit for a detailed VT used in load flow, short-circuit calculation and RMS simulation 9

2.2 Equivalent circuit for a detailed VT used in EMT simulation 13

2.3 Equivalent circuit for a capacitive VT used in load flow, short-circuit calculation and RMS simulation 18

2.4 Equivalent circuit for a capacitive VT used in EMT simulation 24

List of Tables

2.1 Signals, ideal 2- or 3-phase VT, balanced load flow, short-circuit calculation and RMS simulation . . . 6

2.2 Signals, ideal 2- or 3-phase VT, unbalanced load flow, short-circuit calculation and RMS simulation . . . 7

2.3 Calculation parameters, ideal 2- or 3-phase VT, load flow, short-circuit calculation and RMS simulation . . . 7

2.4 Signals, ideal 2- or 3-phase VT, EMT simulation . . . 8

2.5 Calculation parameters, ideal 2- or 3-phase VT, EMT simulation . . . 8

2.6 Calculation parameters, detailed 2- or 3-phase VT, load flow, short-circuit calculation and RMS simulation . . . 12

2.7 Signals, detailed 2- or 3-phase VT, EMT simulation . . . 16

2.8 Calculation parameters, detailed 2- or 3-phase VT, EMT simulation . . . 16

2.9 Calculation parameters, capacitive 2- or 3-phase VT, load flow, short-circuit calculation and RMS simulation . . . 22

2.10 Signals, capacitive 2- or 3-phase VT, EMT simulation . . . 28

2.11 Calculation parameters, detailed 2- or 3-phase VT, EMT simulation . . . 29

2.12 Signals, single-phase VT, balanced load flow, short-circuit calculation and RMS simulation . . . 30

2.13 Signals, ideal single-phase VT, unbalanced load flow, short-circuit calculation and RMS simulation . . . 31

2.14 Calculation parameters, ideal single-phase VT, load flow, short-circuit calculation and RMS simulation . . . 31

2.15 Signals, ideal single-phase VT, EMT simulation . . . 32